

High order method for Black Scholes PDE

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May 1, 2025



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Black-Scholes PDE

- ❑ Black-Scholes PDE is widely used in financial industry to price option, which is a financial derivative with nonlinear payoff.

$$\frac{\partial V(S, t)}{\partial t} = \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V(S, t)}{\partial S^2} + rS \frac{\partial V(S, t)}{\partial S} - rV(S, t)$$

- ❑ For call option:

- Initial condition:
 - $V(S, 0) = \max(S - K, 0)$
- Boundary conditions:
 - $V(0, t) = 0$
 - $\lim_{S \rightarrow +\infty} V(S, t) = S$

- ❑ For put option:

- Initial condition:
 - $V(S, 0) = \max(K - S, 0)$
- Boundary conditions:
 - $V(0, t) = Ke^{-rt}$
 - $\lim_{S \rightarrow +\infty} V(S, t) = 0$

- ❑ Grids: S is stock price, t is time to maturity.

- ❑ Given parameters: σ is volatility of stock, r is risk-free rate and K is strike price.

Space discretization

□ Fourth-order central finite difference

- For an interior grid point, the first derivative:

$$\frac{\partial V_i}{\partial S} = \frac{-V_{i+2} + 8V_{i+1} - 8V_{i-1} + V_{i-2}}{12\Delta S}, 2 \leq i \leq N - 2.$$

- The formula above cannot provide approximation for the first point V_1 and the last point V_{N-1} .
- Use unilateral derivative method:

$$\frac{\partial V_1}{\partial S} = \frac{-3V_0 - 10V_1 + 18V_2 - 6V_3 + V_4}{12\Delta S}$$

$$\frac{\partial V_{N-1}}{\partial S} = \frac{-3V_N - 10V_{N-1} + 18V_{N-2} - 6V_{N-3} + V_{N-4}}{12\Delta S}$$

Space discretization

□ Fourth-order central finite difference

- In the same way, we get formulae for second derivative:

$$\frac{\partial^2 V_i}{\partial S^2} = \frac{-V_{i+2} + 16V_{i+1} - 30V_i + 16V_{i-1} - V_{i-2}}{12\Delta S^2}, 2 \leq i \leq N-2.$$

$$\frac{\partial^2 V_1}{\partial S^2} = \frac{10V_0 - 15V_1 - 4V_2 + 14V_3 - 6V_4 + V_5}{12\Delta S^2}$$

$$\frac{\partial^2 V_{N-1}}{\partial S^2} = \frac{10V_N - 15V_{N-1} - 4V_{N-2} + 14V_{N-3} - 6V_{N-4} + V_{N-5}}{12\Delta S^2}$$

Space discretization

□ Fourth-order central finite difference in matrix form:

$$\frac{\partial V}{\partial S} = AV + b_1$$

$$A = \begin{bmatrix} \frac{-10}{12h} & \frac{18}{12h} & \frac{-6}{12h} & \frac{1}{12h} & & \\ \frac{-8}{12h} & 0 & \frac{8}{12h} & \frac{-1}{12h} & & \\ \frac{1}{12h} & \frac{-8}{12h} & 0 & \frac{8}{12h} & \ddots & \\ & & \ddots & \ddots & \ddots & \\ & & & \frac{1}{12h} & \ddots & 0 \\ & & & \frac{1}{12h} & \frac{-6}{12h} & \frac{18}{12h} \\ & & & & \frac{-10}{12h} & \frac{-3V_N}{12h} \end{bmatrix}$$

$$b_1 = \left(\frac{-3V_0}{12h} \quad \frac{V_0}{12h} \quad 0 \quad \dots \quad \frac{-V_N}{12h} \quad \frac{-3V_N}{12h} \right)^T$$

$$\frac{\partial^2 V}{\partial S^2} = BV + b_2$$

$$B = \begin{bmatrix} \frac{-15}{12h^2} & \frac{-4}{12h^2} & \frac{14}{12h^2} & \frac{-6}{12h^2} & \frac{1}{12h^2} & \\ \frac{16}{12h^2} & \frac{-30}{12h^2} & \frac{16}{12h^2} & \frac{-1}{12h^2} & & \\ \frac{-1}{12h^2} & \frac{16}{12h^2} & \frac{-30}{12h^2} & \frac{16}{12h^2} & \ddots & \\ & & \ddots & \ddots & \ddots & \\ & & & \frac{-1}{12h^2} & \ddots & \frac{-30}{12h^2} \\ & & & \frac{1}{12h^2} & \frac{-6}{12h^2} & \frac{14}{12h^2} \\ & & & & \frac{-4}{12h^2} & \frac{-15}{12h^2} \end{bmatrix}$$

$$b_2 = \left(\frac{10V_0}{12h^2} \quad \frac{-V_0}{12h^2} \quad 0 \quad \dots \quad \frac{-V_N}{12h^2} \quad \frac{10V_N}{12h^2} \right)^T$$

Space discretization

▣ Fourth-order central finite difference

- Now we can rewrite the Black–Scholes PDE in the form of linear system:

$$\frac{\partial V}{\partial t} = PV + Q$$

that,

$$P = \frac{1}{2}\sigma^2 S^2 B + rSA - rI,$$

$$Q = \frac{1}{2}\sigma^2 S^2 b_2 + rSb_1,$$

and I is an identity matrix.

Space discretization

▣ Fourth-order compact finite difference

- The fourth-order central difference method requires the function value on five points nearby at least, and the five-diagonal matrices make the stencil quite huge.
- Use function values and the lower order derivative values on nearby points to approximate higher order derivative values.

$$\frac{\partial V}{\partial S} = F^{-1}GV = DV, \quad \frac{\partial V^2}{\partial S^2} = U^{-1}WV = D^2V$$

- Rewrite the Black–Scholes PDE in the form of $\frac{\partial V}{\partial t} = LV$ where $L = \frac{1}{2}\sigma^2 S^2 D^2 + rSD - rI$.
- It has simpler computing stencil and there is no need to deal with the boundary conditions.

Compact form

$$F = \begin{bmatrix} 1 & 3 & 0 & & & \\ \frac{1}{4} & 1 & \frac{1}{4} & & & \\ & \frac{1}{4} & 1 & \ddots & & \\ & & \ddots & \ddots & \ddots & \\ & & & \ddots & 1 & \frac{1}{4} \\ & & & & 3 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 10 & 0 & & & \\ \frac{1}{10} & 1 & \frac{1}{10} & & & \\ & \frac{1}{10} & 1 & \ddots & & \\ & & \ddots & \ddots & \ddots & \\ & & & \ddots & 1 & \frac{1}{10} \\ & & & & 10 & 1 \end{bmatrix}$$

$$G = \begin{bmatrix} \frac{-17}{6h} & \frac{3}{2h} & \frac{3}{2h} & \frac{-1}{6h} & & & \\ & \frac{-3}{4h} & 0 & \frac{3}{4h} & & & \\ & & \frac{-3}{4h} & 0 & \ddots & & \\ & & & \ddots & \ddots & \ddots & \\ & & & & \ddots & 0 & \frac{3}{4h} \\ & & & & & \frac{1}{6h} & \frac{-3}{2h} & \frac{-3}{2h} & \frac{17}{6h} \end{bmatrix}$$

$$W = \begin{bmatrix} \frac{145}{12h^2} & \frac{-76}{3h^2} & \frac{29}{2h^2} & \frac{-4}{3h^2} & \frac{1}{12h^2} & & & \\ & \frac{6}{5h^2} & \frac{-12}{5h^2} & \frac{6}{5h^2} & & & & \\ & & \frac{6}{5h^2} & \frac{-12}{5h^2} & \frac{6}{5h^2} & & & \\ & & & \frac{6}{5h^2} & \frac{-12}{5h^2} & \frac{6}{5h^2} & & \\ & & & & \frac{6}{5h^2} & \frac{-12}{5h^2} & \ddots & \\ & & & & & \ddots & \ddots & \ddots \\ & & & & & & \ddots & \frac{-12}{5h^2} & \frac{6}{5h^2} \\ & & & & & & & \frac{1}{12h^2} & \frac{-4}{3h^2} & \frac{29}{2h^2} & \frac{-76}{3h^2} & \frac{145}{12h^2} \end{bmatrix}$$

Time discretization

□ Crank-Nicolson method

- The weakness of the explicit and implicit method is that the order of the global truncation error in time is $O(\delta)$, which requires that the time step is much smaller than the spatial step size.
- An effective solution is to average the forward difference method and the backward difference method. It makes the global truncation error order in time $O(\delta^2)$.

Time discretization

□ Fourth-order backward differentiation (BDF4)

- For a general linear system,

$$\frac{\partial V}{\partial t} = AV + b$$

- The form of BDF4,

$$\left(\frac{25}{12}I - \Delta t A\right) V^{j+1} = 4V^j - 3V^{j-1} + \frac{4}{3}V^{j-2} - \frac{1}{4}V^{j-3} + \Delta t b^{j+1}$$

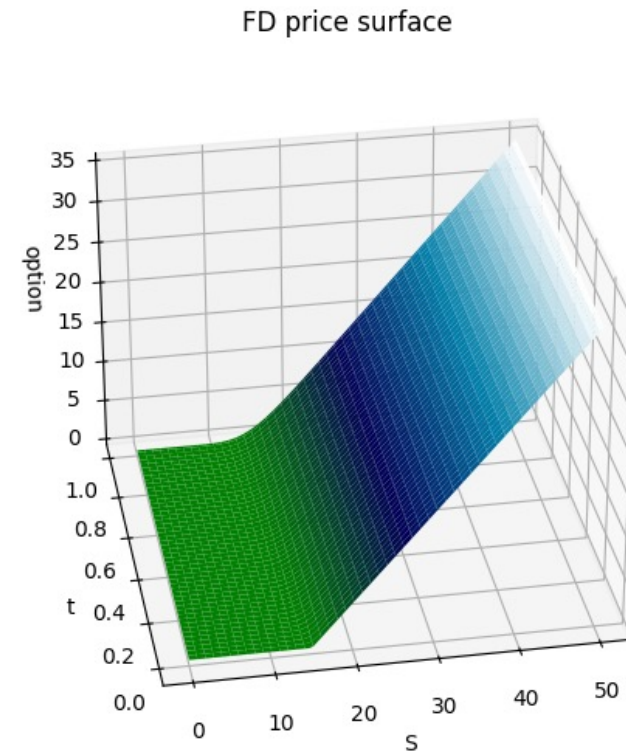
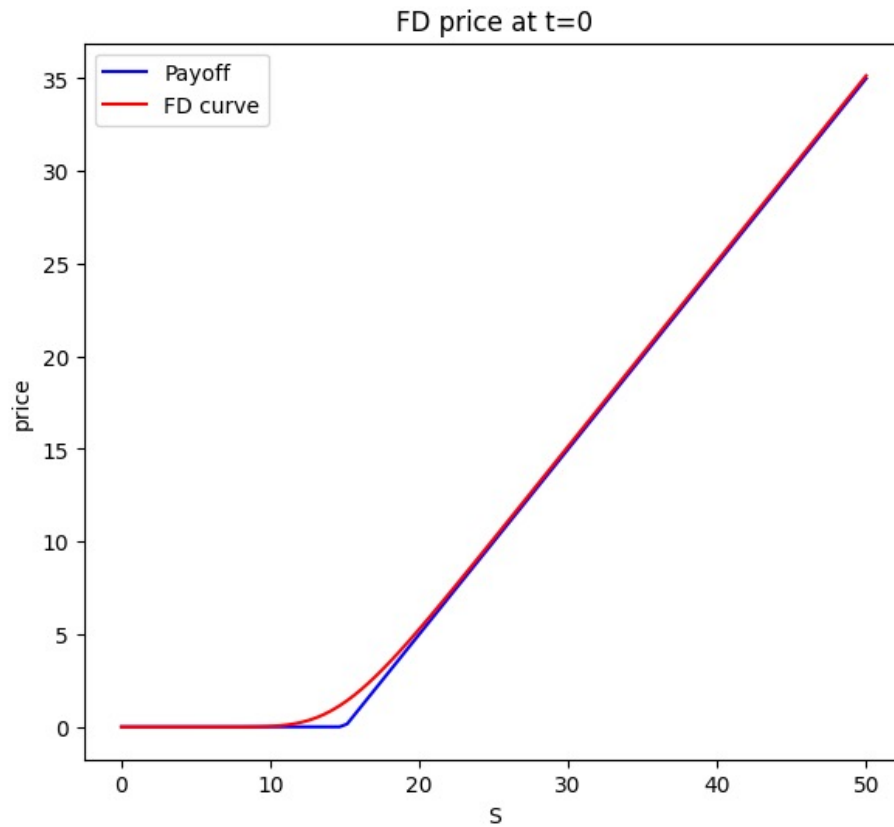
□ BDF4 requires the first 4 initial values.

- A simple way is to use the Crank–Nicolson method twice to obtain V^1 and V^2 from V^0 and then use the BDF3 to obtain the value of V^3 , and finally the value of V^4 is obtained by BDF4.
- The form of BDF3,

$$\left(\frac{11}{6}I - \Delta t A\right) V^{j+1} = 3V^j - \frac{3}{2}V^{j-1} + \frac{1}{3}V^{j-2} + \Delta t b^{j+1}$$

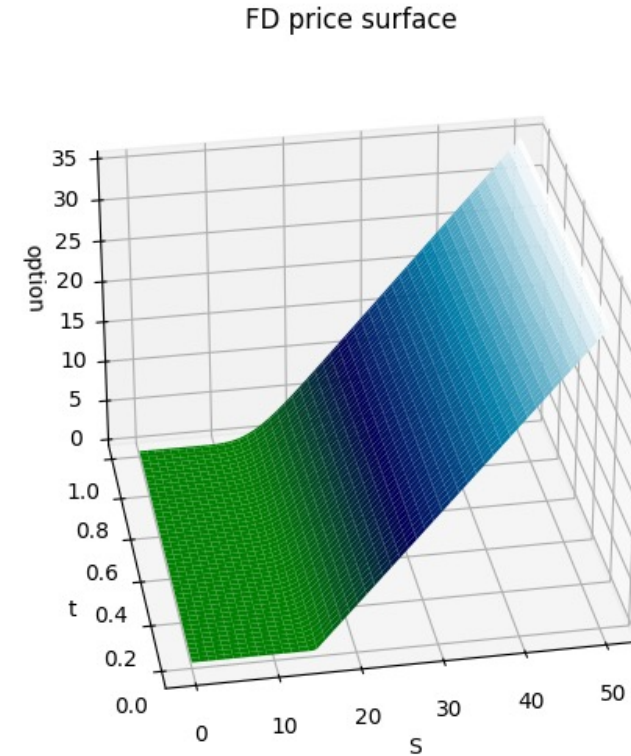
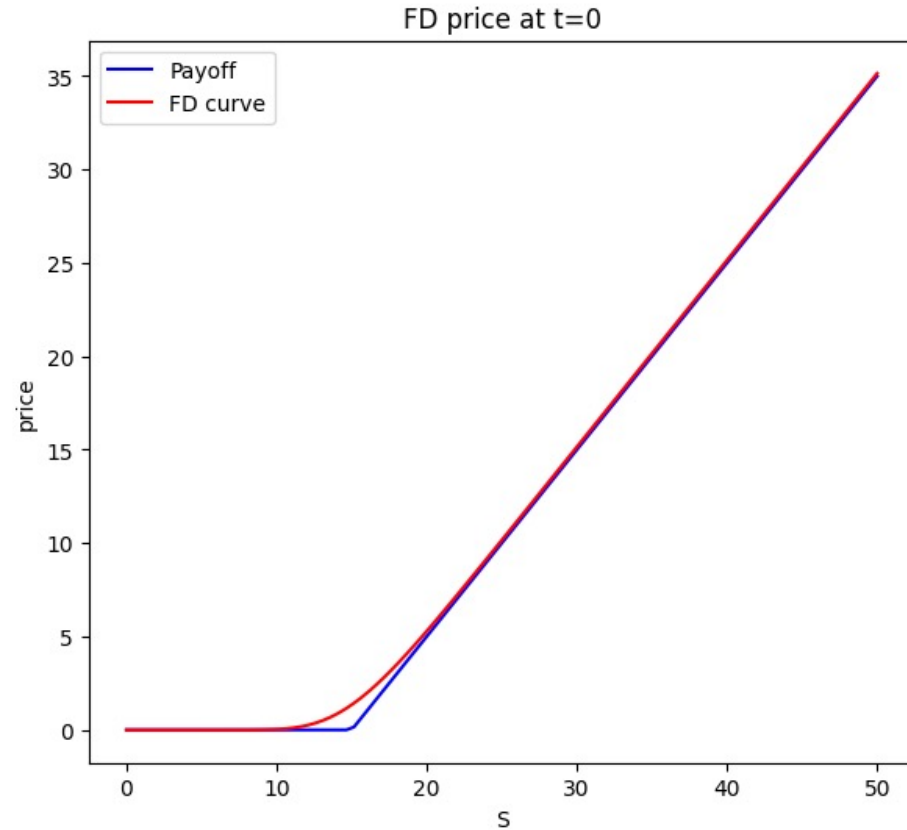
Numerical Results

□ Crank-Nicolson



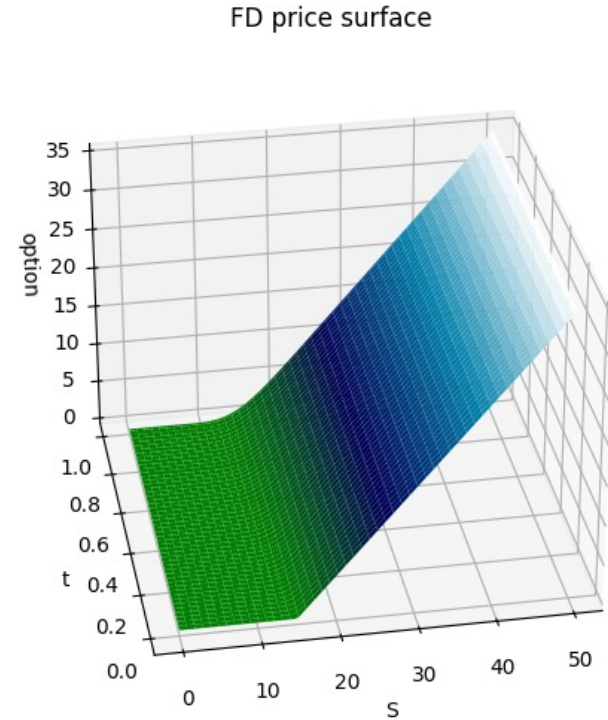
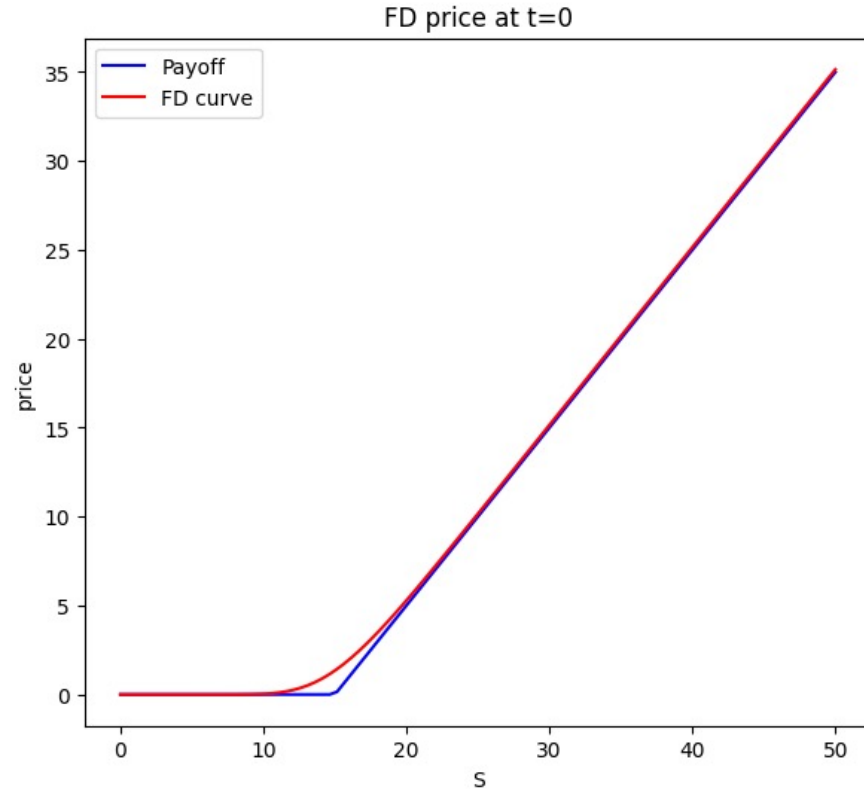
Numerical Results

□ Central BDF4



Numerical Results

□ Compact BDF4



Numerical Results

□ Computational time

Grid	Crank-Nicolson	Central BDF4	Compact BDF4
10x10	0.4218	0.3544	0.3424
20x20	0.0791	0.0558	0.0534
40x40	0.0209	0.0146	0.0144
80x80	0.0048	0.0033	0.0032

□ Truncation error

Grid	Crank-Nicolson	BDF4	Compact BDF4
10x10	-	-	-
20x20	2.4147	2.6679	2.6820
40x40	1.9175	1.9383	1.8934
80x80	2.1380	2.1604	2.1461

□ Crank-Nicolson method is the fastest, but it also has the largest truncation error.

□ In option pricing, the estimate should be accurate up to cent, this means that Crank-Nicolson is enough to handle it with a 80 by 80 grid.

Numerical Results

- ❑ Though we apply fourth-order methods both in space and time, the results don't show fourth-order accuracy.
 - In the option pricing, the initial condition (i.e. the payoff function of the option) is non-differentiable at the strike price K .
- ❑ Further work
 - Apply the non-uniform grid to distribute more points in the vicinity of the bad point (strike price), so that the value of the derivative around the non-differentiable point can be approximated by a smaller step size.

Coordinate transformation

Use coordinate transformation as follows:

$$y = \psi(S) = \frac{\sinh^{-1}(\xi(S - X)) - c_1}{c_2 - c_1},$$

where $c_1 = \sinh^{-1}(\xi(S_{\min} - X))$, $c_2 = \sinh^{-1}(\xi(S_{\max} - X))$ are both constants and parameter ξ is called stretching factor. It is not difficult to verify that this transformation maps $S \in [S_{\min}, S_{\max}]$ on original coordinate system into $y \in [0, 1]$ on new coordinate system. The inverse transformation is

$$S = \psi^{-1}(y) = \varphi(y) = \frac{1}{\xi} \sinh(c_2 y + c_1(1 - y)) + X.$$

Coordinate transformation

After the coordinate transformation, the coefficients also change and can be obtained by the chain rule

$$\frac{\partial V}{\partial S} = \frac{\partial V}{\partial y} \frac{dy}{dS} = \frac{\partial V}{\partial y} \left(\frac{dS}{dy} \right)^{-1} = \frac{1}{\varphi'(y)} \frac{\partial V}{\partial y}$$
$$\frac{\partial^2 V}{\partial S^2} = \left(\frac{dS}{dy} \right)^{-1} \frac{\partial}{\partial y} \left(\left(\frac{dS}{dy} \right)^{-1} \frac{\partial V}{\partial y} \right) = \frac{1}{(\varphi'(y))^2} \frac{\partial^2 V}{\partial y^2} - \frac{\varphi''(y)}{(\varphi'(y))^3} \frac{\partial V}{\partial y}.$$

Thus the transformed Black-Scholes PDE is

$$\frac{\partial V}{\partial t} = \frac{1}{2} \sigma^2 \frac{\varphi(y)^2}{J(y)^2} \frac{\partial^2 V}{\partial y^2} + \left(r \frac{\varphi(y)}{J(y)} - \frac{1}{2} \sigma^2 \frac{\varphi(y)^2}{J(y)^3} H(y) \right) \frac{\partial V}{\partial y} - rV,$$

where $J(y) = \varphi'(y)$ and $H(y) = \varphi''(y)$.

Coordinate transformation

- High order methods can be accurate up to cent in a smaller grid than Crank-Nicolson method.

	Scheme	Grid	Error	Order	Computing time
0	Crank–Nicolson	10 × 10	0.3937	–	0.0012 s
1		20 × 20	0.0729	2.4328	0.0010 s
2		40 × 40	0.0174	2.0667	0.0013 s
3		80 × 80	0.0052	1.7508	0.0023 s
4	BDF4	10 × 10	0.4296	–	0.0039 s
5		20 × 20	0.0089	5.5942	0.0054 s
6		40 × 40	9.6510e–04	3.2040	0.0053 s
7		80 × 80	1.0542e–04	3.1946	0.0083 s
8	Compact BDF4	10 × 10	0.2042	–	0.0033 s
9		20 × 20	0.0103	4.3111	0.0038 s
10		40 × 40	9.4792e–04	3.4401	0.0057 s
11		80 × 80	1.2304e–04	2.9456	0.0067 s

Application

- High order methods calculate more accurate Greeks than Crank-Nicolson method.

Crank-Nicolson:

Greek	FD Value	Analytical	Error
Delta	-0.439674	-0.439118	5.557919e-04
Gamma	0.123652	0.123913	2.607451e-04
Theta	-1.108229	-1.099130	9.098842e-03

BDF4:

Greek	FD Value	Analytical	Error
Delta	-0.439128	-0.439118	1.067707e-05
Gamma	0.123942	0.123913	2.966204e-05
Theta	-1.103333	-1.099130	4.203551e-03

Compact BDF4:

Greek	FD Value	Analytical	Error
Delta	-0.439107	-0.439118	1.103974e-05
Gamma	0.123926	0.123913	1.305129e-05
Theta	-1.103266	-1.099130	4.136128e-03

Reference

- [1] Hu, Jinhao, and Siqing Gan. "High order method for Black–Scholes PDE." *Computers & Mathematics with Applications* 75.7 (2018): 2259-2270.
- [2] Tangman, D. Y., A. Gopaul, and M. Bhuruth. "Numerical pricing of options using high-order compact finite difference schemes." *Journal of Computational and Applied Mathematics* 218.2 (2008): 270-280.